

DDL,DML,DCL,TCL

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SEC : 7

1.DDL COMMANDS : CREATE, ALTER, DROP, TRUNCATE, RENAME :

```
1 • DROP DATABASE DDL;
2 • CREATE DATABASE DDL;
3 • USE DDL;
4
5 • B Mohith Sai Chandra (2520030028)
6 -- DDL COMMANDS - BANK TRANSACTION SYSTEM
7 -- 1. CREATE
8 CREATE TABLE bank_transactions (
9     txn_id INT PRIMARY KEY,
10    customer_name VARCHAR(50),
11    branch_name VARCHAR(50),
12    transaction_type VARCHAR(20),
13    amount DECIMAL(10,2),
14    transaction_date DATE
15 );
16
17 -- 2. ALTER - ADD COLUMN
18 • ALTER TABLE bank_transactions
19 ADD account_no VARCHAR(20);
```

Output

#	Time	Action	Message
✓ 1	19:54:08	DROP DATABASE DDL	1 row(s) affected
✓ 2	19:54:08	CREATE DATABASE DDL	1 row(s) affected
✓ 3	19:54:08	USE DDL	0 row(s) affected
✓ 4	19:54:28	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name...	0 row(s) affected
✓ 5	19:54:49	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0

```

21      -- 3. ALTER - MODIFY COLUMN
22 •    ALTER TABLE bank_transactions
23      MODIFY customer_name VARCHAR(100);
24
25      -- 4. RENAME
26 •    RENAME TABLE bank_transactions
27      TO customer_transactions;
28
29      -- 5. TRUNCATE
30 •    TRUNCATE TABLE customer_transactions;
31

```

Output			
Action Output			
#	Time	Action	Message
✓ 1	19:55:21	ALTER TABLE bank_transactions MODIFY customer_name VARCHAR(100)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
✓ 2	19:55:24	RENAME TABLE bank_transactions TO customer_transactions	0 row(s) affected
✓ 3	19:55:29	TRUNCATE TABLE customer_transactions	0 row(s) affected

```

32      -- 6. DROP
33 •    CREATE TABLE bank_backup (
34          txn_id INT,
35          customer_name VARCHAR(100),
36          branch_name VARCHAR(50),
37          transaction_type VARCHAR(20),
38          amount DECIMAL(10,2),
39          transaction_date DATE,
40          account_no VARCHAR(20)
41      );
42 •    DROP TABLE bank_backup;

```

Output			
Action Output			
#	Time	Action	Message
✓ 1	19:56:02	CREATE TABLE bank_backup (txn_id INT, customer_name VARCHAR(100), ...	0 row(s) affected
✓ 2	19:56:07	DROP TABLE bank_backup	0 row(s) affected

2. DML COMMANDS : INSERT, UPDATE, DELETE, SELECT :

```
43 B Mohith Sai Chandra (2520030028)
44 -- =====
45 -- DML COMMANDS - BANK TRANSACTION SYSTEM
46 -- =====
47
48 DROP TABLE IF EXISTS bank_transactions;
49 CREATE TABLE bank_transactions (
50     txn_id INT PRIMARY KEY,
51     customer_name VARCHAR(50),
52     branch_name VARCHAR(50),
53     transaction_type VARCHAR(20),
54     amount DECIMAL(10,2),
55     transaction_date DATE
56 );
```

Output			
Action Output			
#	Time	Action	Message
1	19:59:15	DROP TABLE IF EXISTS bank_transactions	0 row(s) affected, 1 warning(s): 1051 Unknown table 'ddl.bank_transactions'
2	19:59:25	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name...	0 row(s) affected

```
58 -- 1. INSERT
59 INSERT INTO bank_transactions VALUES
60 (101,'Ravi','Hyderabad','Deposit',5000,'2024-01-05'),
61 (102,'Sita','Hyderabad','Withdrawal',2000,'2024-01-06'),
62 (103,'Kiran','Vijayawada','Deposit',12000,'2024-01-08'),
63 (104,'Anil','Vizag','Deposit',8000,'2024-01-10'),
64 (105,'Priya','Hyderabad','Withdrawal',3500,'2024-01-11'),
65 (106,'Ramesh','Vizag','Deposit',15000,'2024-01-12'),
66 (107,'Keerthi','Vijayawada','Withdrawal',1000,'2024-01-13'),
67 (108,'Rahul','Hyderabad','Deposit',9000,'2024-01-14'),
68 (109,'Sneha','Vizag','Withdrawal',4000,'2024-01-15'),
69 (110,'Madhu','Vijayawada','Deposit',11000,'2024-01-16');
70
71 -- 2. INSERT ONE NEW RECORD
72 INSERT INTO bank_transactions
73 VALUES
74 (111,'Venu','Vizag','Deposit',7000,'2024-01-18');
75
76 -- 3. UPDATE
77 UPDATE bank_transactions
78 SET amount = 5000
79 WHERE txn_id = 105;
```

Output			
Action Output			
#	Time	Action	Message
1	20:00:37	INSERT INTO bank_transactions VALUES (101,'Ravi','Hyderabad','Deposit',5000,'202...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
2	20:00:41	INSERT INTO bank_transactions VALUES (111,'Venu','Vizag','Deposit',7000,'2024-01...	1 row(s) affected
3	20:00:53	UPDATE bank_transactions SET amount = 5000 WHERE txn_id = 105	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

```

81  -- 4. DELETE
82  • DELETE FROM bank_transactions
83  WHERE txn_id = 111;
84
85  -- 5. SELECT ALL
86  • SELECT * FROM bank_transactions;

```

Result Grid   Filter Rows: Edit:    Export/Import:   Wrap Cell Content: 						
	txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
▶	101	Ravi	Hyderabad	Deposit	5000.00	2024-01-05
	102	Sita	Hyderabad	Withdrawal	2000.00	2024-01-06
	103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08
	104	Anil	Vizag	Deposit	8000.00	2024-01-10
	105	Priya	Hyderabad	Withdrawal	5000.00	2024-01-11
	106	Ramesh	Vizag	Deposit	15000.00	2024-01-12
	107	Keerthi	Vijayawada	Withdrawal	1000.00	2024-01-13
	108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14
	109	Sneha	Vizag	Withdrawal	4000.00	2024-01-15
	110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16
*	NULL	NULL	NULL	NULL	NULL	NULL

transactions 1 × 				
Output 				
 Action Output 				
	#	Time	Action	Message
✓	1	20:01:55	DELETE FROM bank_transactions WHERE txn_id = 111	1 row(s) affected
✓	2	20:02:00	SELECT * FROM bank_transactions LIMIT 0, 1000	10 row(s) returned

```

88      -- 6. SELECT DEPOSITS
89      • SELECT * FROM bank_transactions
90      WHERE transaction_type = 'Deposit';
91

```

Result Grid						
		Filter Rows:		Edit:		Export/Import:
						Wrap Cell Content:
	txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
▶	101	Ravi	Hyderabad	Deposit	5000.00	2024-01-05
	103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08
	104	Anil	Vizag	Deposit	8000.00	2024-01-10
	106	Ramesh	Vizag	Deposit	15000.00	2024-01-12
	108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14
	110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16
*	NULL	NULL	NULL	NULL	NULL	NULL

bank_transactions 2 ×

Output



Action Output

#	Time	Action	Message
✓ 1	20:02:50	SELECT * FROM bank_transactions WHERE transaction_type = 'Deposit' LIMIT 0, 1...	6 row(s) returned

```

92 -- 7. ORDER BY
93 • SELECT * FROM bank_transactions
94 ORDER BY amount DESC;

```

Result Grid						
		Filter Rows:		Edit:		
				Export/Import:		
				Wrap Cell Content:		
	txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
▶	106	Ramesh	Vizag	Deposit	15000.00	2024-01-12
	103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08
	110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16
	108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14
	104	Anil	Vizag	Deposit	8000.00	2024-01-10
	101	Ravi	Hyderabad	Deposit	5000.00	2024-01-05
	105	Priya	Hyderabad	Withdrawal	5000.00	2024-01-11
	109	Sneha	Vizag	Withdrawal	4000.00	2024-01-15
	102	Sita	Hyderabad	Withdrawal	2000.00	2024-01-06
	107	Keerthi	Vijayawada	Withdrawal	1000.00	2024-01-13
*	NULL	NULL	NULL	NULL	NULL	NULL

bank_transactions 3 ×			
Output			
Action Output			
#	Time	Action	Message
✓ 1	20:03:15	SELECT * FROM bank_transactions ORDER BY amount DESC LIMIT 0, 1000	10 row(s) returned

3. DCL COMMANDS : GRANT, REVOKE :

```

100 ✖ B Mohith Sai Chandra (2520030028)
101 -- Create table
102 DROP TABLE IF EXISTS bank_transactions;
103 • CREATE TABLE bank_transactions (
104     txn_id INT PRIMARY KEY,
105     customer_name VARCHAR(50),
106     branch_name VARCHAR(50),
107     transaction_type VARCHAR(20),
108     amount DECIMAL(10,2),
109     transaction_date DATE
110 );
111
112 -- CREATE USERS
113 • CREATE USER IF NOT EXISTS 'Auditor1'@'localhost'
114     IDENTIFIED BY 'Auditor@123';
115 • CREATE USER IF NOT EXISTS 'BranchManager'@'localhost'
116     IDENTIFIED BY 'Manager@123';

```

Output			
Action Output			
#	Time	Action	Message
✓ 1	20:05:49	DROP TABLE IF EXISTS bank_transactions	0 row(s) affected
✓ 2	20:05:53	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name...	0 row(s) affected
✓ 3	20:06:02	CREATE USER IF NOT EXISTS 'Auditor1'@'localhost' IDENTIFIED BY 'Auditor@123'	0 row(s) affected
✓ 4	20:06:02	CREATE USER IF NOT EXISTS 'BranchManager'@'localhost' IDENTIFIED BY 'Manager@123'	0 row(s) affected

```

118 -- 1. GRANT SELECT
119 • GRANT SELECT
120 ON bank_transactions
121 TO 'Auditor1'@'localhost';
122
123 -- 2. GRANT ALL
124 • GRANT ALL PRIVILEGES
125 ON bank_transactions
126 TO 'BranchManager'@'localhost';
127
128 -- 3. REVOKE SELECT
129 • REVOKE SELECT
130 ON bank_transactions
131 FROM 'Auditor1'@'localhost';
132
133 -- 4. REVOKE ALL
134 • REVOKE ALL PRIVILEGES
135 ON bank_transactions
136 ❌ FROM 'BranchManager'@'localhost';

```

Output

📄 Action Output ▼

	#	Time	Action	Message
✓	1	20:06:39	GRANT SELECT ON bank_transactions TO 'Auditor1'@'localhost'	0 row(s) affected
✓	2	20:06:45	GRANT ALL PRIVILEGES ON bank_transactions TO 'BranchManager'@'localhost'	0 row(s) affected
✓	3	20:06:50	REVOKE SELECT ON bank_transactions FROM 'Auditor1'@'localhost'	0 row(s) affected
✓	4	20:09:27	REVOKE ALL PRIVILEGES ON ddl.bank_transactions FROM 'BranchManager'@'loca...	0 row(s) affected

4. TCL COMMANDS : COMMIT, SAVEPOINT, ROLLBACK :

```
142 B Mohith Sai Chandra (2520030028)
143 -- Create table
144 DROP TABLE IF EXISTS bank_transactions;
145 CREATE TABLE bank_transactions (
146     txn_id INT PRIMARY KEY,
147     customer_name VARCHAR(50),
148     branch_name VARCHAR(50),
149     transaction_type VARCHAR(20),
150     amount DECIMAL(10,2),
151     transaction_date DATE
152 );
153 -- CREATE USERS
154 CREATE USER IF NOT EXISTS 'Auditor1'@'localhost'
155 IDENTIFIED BY 'Auditor@123';
```

Output

Action Output

#	Time	Action	Message
✓ 1	20:12:52	DROP TABLE IF EXISTS bank_transactions	0 row(s) affected
✓ 2	20:12:56	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name...	0 row(s) affected

```
154 -- INSERT SAMPLE DATA
155 INSERT INTO bank_transactions VALUES
156 (101,'Ravi','Hyderabad','Deposit',5000,'2024-01-05'),
157 (102,'Sita','Hyderabad','Withdrawal',2000,'2024-01-06'),
158 (103,'Kiran','Vijayawada','Deposit',12000,'2024-01-08'),
159 (104,'Anil','Vizag','Deposit',8000,'2024-01-10');
160
161 -- 1. COMMIT
162 START TRANSACTION;
163 UPDATE bank_transactions
164 SET amount = 6000
165 WHERE txn_id = 101;
166 COMMIT;
167
168 -- 2. SAVEPOINT
169 START TRANSACTION;
170 SAVEPOINT Before_Update;
```

Output

Action Output

#	Time	Action	Message
✓ 1	20:13:22	INSERT INTO bank_transactions VALUES (101,'Ravi','Hyderabad','Deposit',5000,'202...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0
✓ 2	20:13:31	START TRANSACTION	0 row(s) affected
✓ 3	20:13:31	UPDATE bank_transactions SET amount = 6000 WHERE txn_id = 101	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 4	20:13:31	COMMIT	0 row(s) affected
✓ 5	20:13:47	START TRANSACTION	0 row(s) affected
✓ 6	20:13:47	SAVEPOINT Before_Update	0 row(s) affected


```
172      -- 3. ROLLBACK TO SAVEPOINT
173 •    UPDATE bank_transactions
174      SET amount = 99999
175      WHERE txn_id = 102;
176 •    ROLLBACK TO SAVEPOINT Before_Update;
```

Output

 Action Output 

	#	Time	Action	Message
✓	1	20:14:25	UPDATE bank_transactions SET amount = 99999 WHERE txn_id = 102	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓	2	20:14:26	ROLLBACK TO SAVEPOINT Before_Update	0 row(s) affected